

Aniruddh Ballal

+91-9731006863 aniruddhballalportfolio.vercel.app Bengaluru, Karnataka, PIN - 560094

Summary

Systems-oriented engineer focused on building real-world applications grounded in underlying models — from scalable full-stack systems to physical simulation and financial modelling. Strong in end-to-end development, high-performance computing, and debugging complex systems, with a consistent bias toward understanding how things truly behave rather than just making them work. Currently exploring the intersection of engineering, physics, and finance.

Education

Manipal Institute of Technology Manipal, KA
B.Tech in Information Technology | CGPA: 8.01 *Expected May 2026*

Deeksha Integrated PU College Yelahanka, KA
Karnataka 2nd PUC Board Examination: 94.2% *May 2022*

Daffodils English School Sanjaynagar, KA
ICSE 10th Board Examination: 92.9% *May 2020*

Experience

Data Analyst Intern | Qwixpert Consulting Jan 2026 – Mar 2026

- Built and deployed end-to-end full-stack internal tools, translating client data and legacy Excel workflows into scalable, high-performance solutions; diagnosed real-world problems and delivered accurate, maintainable systems.

Frontend Web Dev Intern | Indiassetz Jun 2025 – Jul 2025

- Collaborated with design and engineering teams to enhance UI/UX using React and Material UI; implemented micro-interactions and component redesigns following best practices for clean, reusable, version-controlled code.

Solar Physics Intern | Department of Physics, IISc Jun 2024 – Aug 2024

- Worked under Dr. Prantika Bhowmik on a Python-based project modelling and visualising solar magnetic field evolution and cyclical patterns using spherical harmonics, SciPy, AstroPy, NumPy, and Matplotlib; surfaced new quantitative insights into the Sun's magnetic cycle behaviour.

Junior Avionics Member | ThrustMIT Nov 2022 – Nov 2023

- Designed and built avionics systems for a student rocketry team: custom PCBs, telemetry, RF/GPS/IMU module integration with microcontrollers, control theory, embedded programming, and delivery of a hands-on rocketry workshop.

Projects

Sunalyse *React, TypeScript, Three.js, Python, SciPy, PFSS, GLSL* 2025–2026

- Interactive 3D solar coronal magnetic field viewer spanning 181 Carrington Rotations (2010–2024) — a full solar cycle. Python PFSS pipeline computes spherical harmonic coefficients (lmax=85) from NASA/SDO HMI FITS magnetograms; 500 adaptive seeds placed at peak photospheric field locations via haversine clustering; field lines traced with Euler integration across 4 CPU cores on Kaggle (˜20 hours total compute).
- Three.js + React frontend renders three physics layers: colour-mapped photosphere texture, 3D coronal field lines colour-coded per-vertex by field strength, and the heliospheric current sheet at 2.5 Rs via a custom GLSL fragment shader sampling a Float32 spherical texture.

Queen's Gambit 2.0 *React, TypeScript, Rust, WebAssembly, Web Workers, AES-256-GCM* 2025–2026

- Rebuilt a chess-based steganographic file encoder as a fully client-side web app. Replaced chess.js with a Rust move generator (shakmaty) compiled to WebAssembly via wasm-pack; encode/decode loops run in Web Workers with zero-copy ArrayBuffer transfer — achieving a **˜38x encryption and ˜37x decryption speedup** over the original Python monolith.
- AES-256-GCM encryption with PBKDF2 key derivation via the Web Crypto API; random IV ensures distinct chess move sequences per run. No file data ever leaves the device. Backed by a Node.js + Express + MongoDB analytics service.

Credex *Next.js, Node.js, MongoDB, Python, FastAPI, XGBoost, PuLP* 2026

- Unified VenPay, LOANlo, and VenPayRL into a single B2B procurement and trade finance platform for Indian SMEs across 17 MongoDB collections with immutable payment and audit records. Buyers submit RFQs, negotiate terms, confirm POs, and access working capital within one product.
- FastAPI microservice runs a PuLP/CBC cash flow optimiser (90-day horizon) and an XGBoost credit scoring model producing a 300–900 composite score. KYC documents stored in Backblaze B2 with signed 15-minute expiring URLs and an immutable access log per view.

LOANlo *React, TypeScript, Node.js, MongoDB, GridFS, Python, Gmail OAuth2* 2025

- Role-based loan origination platform for applicants, underwriters, and admins. Applicants complete gated multi-step onboarding and track applications through a full review lifecycle; underwriters manage loan products; admins govern soft-deleted records, restoration requests, and IP whitelisting.
- Security: CAPTCHA on auth, rate limiting, input sanitisation, inactivity-based auto logout, session tracking with device/IP metadata. Gmail OAuth2 drives transactional email at every lifecycle event. Python ML microservice handles credit-risk prediction.

VenPay *React, TypeScript, Node.js, MongoDB, Razorpay, Cloudinary, Redux* 2025

- Full-stack B2B procurement platform handling the complete vendor-company transaction lifecycle: listings, purchase requests with Net30/custom deadlines, Razorpay Checkout with partial payment support, real-time transaction history, and post-settlement ratings unlocking after full payment clears.

SatLies *React, TypeScript, Three.js, Python, FastAPI, NumPy, SciPy* 2026

- Satellite mission simulator backed by real astrodynamics: RK45 orbit integrator with optional J2 oblateness perturbation on a 3D Earth, Hohmann transfer mission planner (delta-V budget, Tsiolkovsky fuel mass), and ground station pass predictor (next-pass time, duration, max elevation).

Community & Leadership

Managing Committee Member | The Astronomy Club, MIT Manipal Feb 2024 – Apr 2025

- Completed coursework in History & Philosophy of Astronomy, AstroTech, Black Holes, Astrobiology, and Data-Driven Astronomy; active role in organising club events and outreach.

Organising Committee — Painting & Publicity | Revels, MIT Manipal Jan 2024 – Mar 2024

- Led a sub-team of 4 to paint 10+ acrylic posters ($14' \times 8'$ each) and craft 3D models for MIT's annual cultural festival drawing 25,000+ footfalls.

Working Committee Member | Manipal Runners' Club Dec 2023 – Feb 2024

- Co-organised the Manipal Marathon 2024 (8,000+ participants); personally coached a group of enrollees with structured training over 6 weeks leading up to the event.

Skills

Languages: TypeScript, Python, Rust, JavaScript, C/C++, Java, SQL

Frontend: React, Next.js, Three.js, React Native, Tailwind CSS, Framer Motion, GLSL

Backend & Data: Node.js, Express, FastAPI, MongoDB, PostgreSQL, GridFS, Redis

ML / Systems: TensorFlow.js, XGBoost, PuLP, SciPy, NumPy, WebAssembly, Web Workers

Tools: Git, Docker, Vercel, Cloudinary, Postman, Arduino IDE, KiCad

Interests: Astronomy, Endurance Athletics, Avionics, Cinema